



Plotting Extrapolated Data using Curve Fitting Transcript

Hello everyone, welcome to bitsplanet digital. Let's continue our learning journey for the course data visualisation storytelling. Today's class will be understanding how to fit functional forms to the underlying data.

What do I mean by this? Generally in the data visualisation process often you don't know what is the nature of the underlying data, what is the trend that is there in the data and what shape it can take in the future. For example, you are in a setting where you are trying to predict sales for the next year or you are trying to understand the cash flows for the next financial year. In this case you are expected to predict the data for the next time period or into the future.

The way you generally approach this problem is to understand the existing trend in the data and try to extrapolate it to the next time period. So often this is where modelling comes into play in understanding which particular functional form best represents the underlying data. So let us now try out to understand what is the underlying data form and how best we can use certain functions to fit that data.

In this process you would like to visualise the data first and then understand whether it is performing in I mean whether there is a linear trend whether there is an exponential trend or it is moving in a cyclical pattern like that. For example, in the last class we have seen the temperature is following a cyclical pattern so we can simply see okay during certain months there will be peak temperature of 35 degrees centigrade followed by a lower temperature so on and so forth. So this is a trend and in this case this is looking like a sine wave and you can use a sine wave function to predict this or extrapolate this and there are there can be more complex functions but the very fact you are trying to come to a conclusion about the underlying trend is by visualising the data.

So let us now start by visualising the data and then fitting a function to extrapolate that data and see how it can be used for forecasting purposes. So now let us start with understanding the functions that are there in Python environment that help you in doing this particular fitting. So this is called plotting extrapolated data that you are extrapolating data from the current time period to the next time period using a particular function.

Mind you when you are extrapolating data you are actually creating data that is in the future expecting a particular functional form based on the current data. Always remember there is scope for error in this exercise and the more accurate your functional form is to the real data more accurate will be your forecast. And again this is an earth there is no steadfast rule that particular function fits better to a given data it might not fit well to the some other data set.

Now let us see how you can achieve this task. The task is two parts first is generating the expropriated data and then using a curve fitting to show how the data will look in the future. So you can model your existing data to predict and visualise future trends that's the key essence of today's lecture and we'll see what functions in Python will enable you to do that while also visualising the same.

Now what are we trying to do we are trying to fit a polynomial curve right polynomial means which has multiple variables polynomial and the polynomial is determined by the degree so if it is linear in degree 1 you just have a linear function if it is degree 2 you have a parabolic then cube so on and so forth. So we are using the polynomial function to fit the data and then we get the polynomial coefficients so that we can estimate the relationship or establish the relationship okay. So our interest is not particularly in this function but to see visually whether this function is capturing the reality or the trend in the underlying data.

So now let us start with a small example so what are we doing we are simply running the we are trying to get two variables the one variable is the months they are arranged from 1 to 30 that is 12 months okay it is the date of active monthly users may be on a website or a gaming platform whatever might be the use case that we have right or number of footfalls that you have in the firm you have monthly footfalls or monthly active users for your website so you have arranged them so each one corresponds to the number of users in that particular month and you have the users both are arranged in this particular format so you are just passing these two variables into the Colab environment so you have done that then subsequently you are trying to determine the coefficients right from the polynomial fit function so we are determining a function called coefficients by using `np.polyfit` which fits a polynomial function to this particular two variables and you are using second degree polynomial which is nothing but a parabola right which looks like a parabola okay y is equal to x square and the function so you have this months and users months and then users so then you have the p is equal to polynomial coefficients so you get the coefficients here from the polynomial function

so first you have to determine the part you have to fit the polynomial function and then compute the coefficients so you are doing that part also and just running the selection right then you have the future months so now you are coming to the step where you are trying to predict into the future right so you have 13 months data 12 months data now you want to extend it to 18 months so you are plotting from 1 to 19 months you are constructing the entire curve from 1 to 19 months and you are using the coefficients that you have generated from the fit so the actual logic is you have the polynomial function you have the data and you are fitting the polynomial function and then using those coefficients you are trying to plot the polynomial function for the future months so I have just selected 19 months now what am I doing I am using the figure to showcase the different paths one is the actual path then is the extrapolated trend so figure size I'm just plotting the figure size with all this so I just got the figure right then

I am plotting first line okay which is nothing but months users and it is the marker it is simply the actual monthly users then I am plotting future months which is this and future users will come from the predicted function which uses the coefficients from the polynomial function based on the future months and then line with these two you are using instead of marker which is O you are using double hash then you are using dark orange and labelling it as extrapolated trend and all these are the legend items so I'm just running the entire selection again so this is what I'm looking so the way that you understand is you have this blue dots which are the original data right which are coming from monthly users 1 2 3 4 up to 12 months then you are extending the data to 19 months or 18 months and how is this line is created the line is created based on fitting a polynomial function to the original data and getting the coefficients right coefficients is equal to $n \cdot \text{poly flip}$

And then passing them to a function right poly by first difference right a first degree polynomial of first degree sorry then you are trying to again use the future months which are based on the coefficients that you got from the polynomial feed and showing so this nicely shows that the data is almost captured by the polynomial function but I just want you to have a close attention on this particular area where there is some amount of difference in terms of the fitting it is not passing exactly it is actually depicting almost close behaviour let me showcase this with a another counter example so that you will understand why fitting the right function is important now in this case what am I doing I am plot instead of using a polynomial function I am plotting a linear function so we are not getting to the exact technicalities which we will see a bit later but we are just doing a linear regression function here so I'll just explain the code this scikit-learn library we are getting linear regression model and then the plot function and the end and umpire pandas but you have to arrange rearrange them structurally so that it can be used in scikit-learn so we are just doing the same both months and users and we are using the linear regression model and we are using months and users as a linear regression okay the dependence is linear not polynomial and we have predicted the future months using the linear model the model that whatever we have we are predicting the future future months using the regression coefficients from the linear regression and then rest is all the same and then we plotted the graph so what is happening once you plot the linear this is the linear trend which is showing that on 18 months your users will be thousand right and

but you can see the function is not exactly or not more approximately fitting the data there are some gaps here and there as compared to polynomial function where the predictions are where the graph the predicted graph traces the actual data points more accurately resultantly you have a predicted number of 1400 monthly users as opposed to a linear model which predicts only thousand users which is almost 40 percent difference in the predicted values therefore choosing the right functional form becomes really important in terms of understanding what is the underlying data structure or the trend and use the function for plotting and finally it is sometimes you might need not visualise if you know the exact data structure but visualisation helps us to understand



and most importantly to showcase our audience how close are we tracking the real data and how strong are our predictions so imagine you can run this model without visualising and got a prediction and showed to your management or to your manager but you would be more convinced which is the entire purpose of your visualisation exercise to bring in acceptance much more efficiently from your audience by showcasing the visual with the predicted and actual values you are trying to bring in more credibility to your story saying that as I have predicted sales I have predicted my profits or I predicted the number of users and this is how the data is looking like

And therefore I am using this function I am more confident of this materialising rather than just take my word for it you are showing it visually which will bring in more acceptance okay so again there are multiple ways to fit this particular function you need not just fit a polynomial if the data is behaving differently you might fit a Gaussian distribution or you can do a higher order polynomial depending upon the use case we might see couple of them in the live lecture but I advise you go back to the documentation in matplotlib or other open sources where you can get your get exposed to the different types of exponential fitting functions of the prediction functions.

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